

A Collaboration for Standardizing Physiological Data Quality Assessment in Neuroimaging

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We're Developing This Together – Join the Discussion!



Share your group's approach to physiological QA



Help identify common challenges



Join future discussions, tutorials, or tool development



Stay in the loop with updates and outputs

WHY?

Natural fluctuations in autonomic physiology, such as breathing and heart rate, **provide windows** into critical functions including cognition, emotion, and health [1-7], and can **heavily influence** fMRI signals [8].

Incorporating physiological recordings into fMRI analysis can improve both noise modeling and neuroscientific interpretation.

HOWEVER,

- Signal quality and handling expertise is highly variable across labs,
- Systematic artifact documentation across datasets is a gap that we can fill.
- Community practices are the foundation for developing automated QA and standards.

CURRENT EFFORT

A critical first step toward developing standardized quality assessment pipelines is the systematic documentation and characterization of clean waveforms and artifacts across different datasets, acquisition protocols.

We are curating information (started with our lab) about how different groups approach physiological data QA, including cardiac, respiratory, CO2, eye tracking, and other measures. We are particularly interested in specific protocols, tools, and decision-making processes.

A Collaboration for Understanding Community Practices and Standardizing Physiological Quality Assurance in Neuroimaging

This template is designed to collect feedback on how the neuroimaging and broader community quality assesses physiological data. Please provide as much information as possible in each section, and links to relevant literature.

Insert a section with your lab's name as the title of the section.

Thank you for your time and input!

Section 1: What kind of physiological data do you investigate? (e.g., cardiac, respiratory, CO2, eye tracking)

Section 2: How do you assess your data?

- In which stage you assess the quality of your data, raw or preprocessed or both?
- How do you assess the quality of your data? (e.g., visual inspection, statistical analysis, machine learning)
- What tools or software do you use for data quality assessment? Share links to repos or scripts whenever possible.
- How do you look at your data? (e.g., whole-signal, windowed sections)
- What do you look at in your data? (e.g., peaks, irregularities)
- What specific signal quality index (SQI) metrics do you use?

Section 3: How do you "fix" your data?

Section 4: Which datasets have you already QA'ed?

- List the datasets that you have already quality assessed.
- For each dataset, please provide the following information:
 - Number of samples (available in the dataset, qa'ed, passed quality checks)
 - Measures of variability (e.g., recording device, scanner, age, etc.)

We encourage you to add yours here as well >>> [Contributing Labs Physiological Data QA info](#)

Additionally: We would love to collect some examples of what you deem good, bad and anything in between. Please share if you'd like screenshots with descriptions or spreadsheets with decisions here >>> [QA'ed physiological signals](#)

Section 5: Additional Comments

- Please provide any additional comments or suggestions that you have regarding data quality assessment.

CURRENT COLLABORATORS



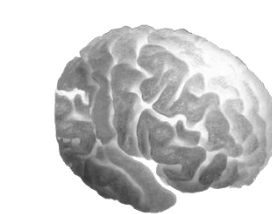
Department of Bioengineering

BiosigsysLab

CANDY Lab



SIMEXP Lab



Université de Montréal

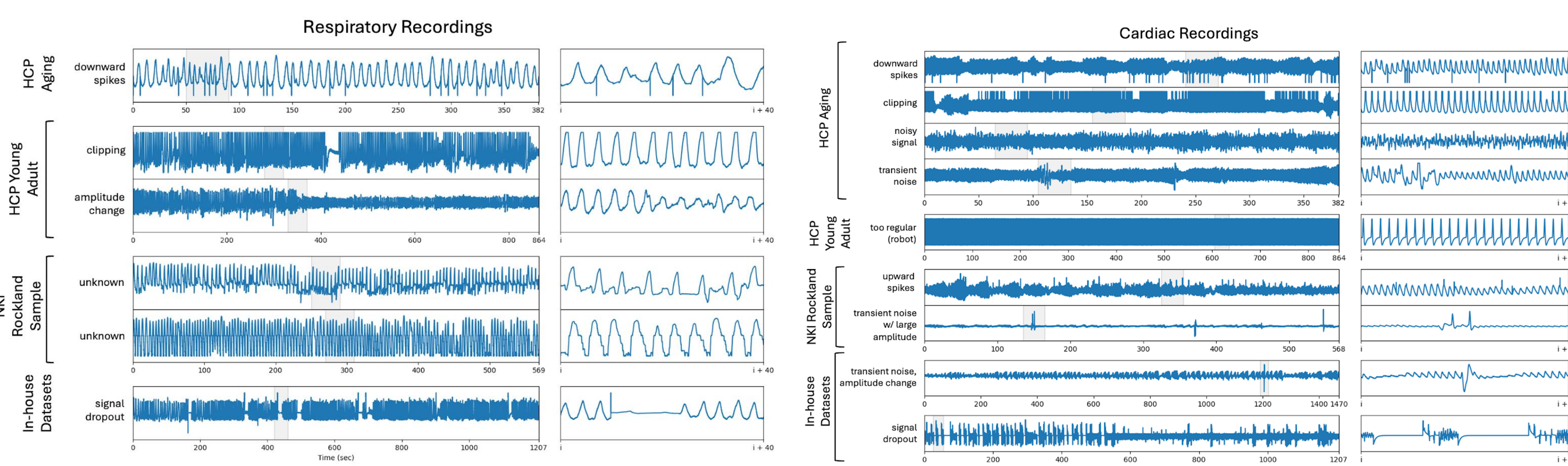
Bright Lab
- ANVIL



WHAT WE HAVE LEARNED SO FAR

We have ...

- Inspected five datasets (3 of which are large public datasets): 5000+ raw physiological recordings by five experts using an in-house manual annotation tool.
- Identified recurrent artifacts in photoplethysmography (PPG) and respiratory belt data, documented their features, and provided annotated examples.
- Documented key acquisition parameters including the types of physiological data collected (respiratory, cardiac, eye tracking), recording devices and any known specifications, experimental tasks, scanner information (e.g., 3T Siemens Prisma), sequence details (e.g., single vs. multi-echo), and participant demographics (age range).



Two key take aways from our explorations and collaborations:

- Most labs manually inspect their data. This inspection is not only dataset but research question specific. Inter-rater variability suggest, QA could benefit from automated tools.
- Unique datasets has unique artifacts. It is worth exploring the underlying cause, whether it is induced by hardware, different scanners, only apparent at certain sampling rates etc.

NEXT STEPS



- A virtual meeting with interested folks after OHBM.
- Publish (open access) widely used practices and guidelines in the community including in-house tools.
- Share research artifacts, i.e. physiological data QA of publicly available datasets.

INTENDED OUTCOMES

Learn from community!
Give back to community!
Maintain an open access, living document!
Improve QA pipelines!
Build collaborations!

REFERENCES

- [1] Yuan et al. (2013), PMID: 23631982
- [2] Barrett and Simmons (2015), PMID: 26016744
- [3] Shokri-Kojori et al. (2018), PMID: 30566618
- [4] Mather et al. (2018), PMID: 29333483
- [5] Azzalini et al. (2019), PMID: 31047813
- [6] Koban et al. (2021), PMID: 33790441
- [7] Shams et al. (2021), PMID: 33516896
- [8] Uddin (2020), PMID: 32600967